Data and Code for: GDP-B: Accounting for the Value of New and Free Goods

Overview

This replication package accompanies the study "GDP-B: Accounting for the Value of New and Free Goods". The study uses survey data to estimate consumer valuation of free digital goods such as Facebook and smartphone cameras. The data were collected through online surveys hosted on https://my.preferencelab.com.

The replication package contains:

- Three survey data files: `Facebook.csv`, `Cameras.csv`, and `Lab_studies.csv`
- Three corresponding R scripts: `Facebook.R`, `Cameras.R`, and `Lab_studies.R`

Each script processes the corresponding dataset, performs the analysis, and generates the tables and figures presented in the main paper and appendix.

- Excel file with calculations: `Excel.xlsx`

The code is written in R version 4.3.1 (2023-06-16) and was last run on a MacBook Air M1 (macOS 15.2, 8 GB RAM). Runtime for the full set of scripts is minimal and completes in a matter of seconds on a standard desktop machine.

Data Availability and Provenance

The data used in this study were collected via online surveys hosted on https://my.preferencelab.com. The datasets include survey data about evaluations of digital

goods and their perceived value to consumers. All data were collected by the authors, and Institutional Review Board (IRB) approval was obtained for all studies.

Three datasets are included:

`Facebook.csv`

Survey data from U.S.-based respondents estimating the willingness-to-accept (WTA) for giving up Facebook for one month. The dataset contains 2,885 observations and includes:

- `ID`: Participant identifier
- `Price`: Dollar amount offered to give up Facebook
- `DC`: Discrete choice (1 = accepted money, 0 = preferred Facebook)
- `gender, age, us.region, edu, hh.income`: Demographics
- `weight`: Survey weight to match U.S. Census distribution (see example code in the script)

`Cameras.csv`

Survey data from students estimating WTA for giving up smartphone camera functionality via a BDM lottery. The dataset contains 213 observations and includes:

- `ID`: Participant identifier
- `Gender`, Age_category: Demographics
- `BDM_minimum.for.giving.up.both.smartphone.cameras.for.1.month`: Bid amount
- `Pictures.per.week, Videos.per.week`: Usage metrics

`Lab_studies.csv`

Survey data from lab participants estimating WTA for several free digital goods. The dataset contains 426 observations and includes:

- `ID`: Participant identifier
- `Age_category, Gender`: Demographics
- `Price_variables`: Offered prices for giving up specific goods
- `DC_variables`: Discrete choice for each good (1 = accepted, 0 = rejected)

A PDF version of the questionnaires is included in the replication archive.

Statement about Rights

The authors certify that they have legitimate access to and permission to use the data included in this replication package. All data were collected by the authors and may be publicly redistributed.

License for Data

The data are licensed under a Creative Commons Attribution-NonCommercial (CC BY-NC) license.

Computational Requirements

- R version: 4.3.1 (2023-06-16)
- Operating system: macOS 15.2 (Apple Silicon M1)
- Memory: 8 GB
- Runtime: <10 minutes total

- Storage: <25 MB
- Required R packages: plyr (listed and loaded within each script)
- Random seeds: Set in each script to ensure reproducibility

Description of Programs / Code

Each script processes its dataset, performs model estimation, and generates relevant figures and statistics.

- `Facebook.R`: Estimates discrete choice models, computes bootstrapped WTA for Facebook, and plots demand curve (Figure 2).
- `Cameras.R`: Computes WTA using BDM bids and plots camera demand curve (Figure 3).
- `Lab_studies.R`: Estimates WTA for various goods, replicates Table A1 and Figure A1 (Online Appendix).

All figures and tables referenced in the corresponding parts of the paper are fully reproducible using these scripts.

Excel Analyses

The Excel file is named "Excel.xlsx" and is self-contained with all data and analyses required to reproduce Table 1 in the main text and Table A2 in the appendix.

Instructions to Replicators

To replicate the results:

1. Place the .csv data files and corresponding .R script in the same directory.

2. Open the script in R (version 4.3.1 or later).
3. Run the script. Outputs (figures, results) will be printed to the console.
4. For the Excel analyses, open the Excel file to see the calculations.

Additional information about the platform used for running the surveys

- Access commitment: We will make the platform for data collection available free of charge for at least five years after publication.
- Study duplication for reproducibility: Anyone wishing to replicate the experiments can be given free access to an exact duplication of the studies on Preference Lab.
- Study IDs for duplication:
    - Cameras: #1100
    - Facebook: #703
    - Lab studies: #994
- Platform version: All studies were conducted using version 1.0 of Preference Lab.